

# Attention based Encoder Decoder Calculations

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## Encoder

| Lets                                       | go                       |                           |              |
|--------------------------------------------|--------------------------|---------------------------|--------------|
| [1.87, 0.09]                               | [-0.78, 0.27]            |                           |              |
| Positional encoding                        |                          |                           |              |
| + [0.0, 1.0]                               | + [-0.9, 0.4]            |                           |              |
| [1.87, 1.09]                               | [-1.68, 0.67] .....(1)   |                           |              |
| Attention calculation in encoder (Lets go) |                          |                           |              |
| Q=1.1 0.6<br>-2.8 2.4                      | K = -1.7 0.5<br>-1.5 0.9 | V = 1.5 -1.0<br>-0.3 -0.2 |              |
| QL = Lets X Q                              | KL = Lets X K            | VL = Lets X V             |              |
| QG = go X Q                                | KG = go X K              | VG = go X V               |              |
| QL = [-1.0, 3.7]                           | KL = [-4.7, 1.9]         | VL = [2.5, -2.1]          |              |
| QG = [-3.7, 0.6]                           | KG = [1.9, -0.2]         | VG = [-2.7, 1.5]          |              |
| QLxKL = 11.7                               | QLxKG = -2.6             | QGxKL = 18.5              | QGxKG = -7.2 |
| Softmax(11.7, -2.6)                        | Softmax(18.5, -7.2)      |                           |              |
| 1.0, 0.0                                   | 1.0, 0.0                 |                           |              |
| VLx1.0 + VGx0                              | VLx1.0 + VGx0.0          |                           |              |
| [2.5, -2.1]                                | [2.5, -2.1]              |                           |              |

From (1) [1.87, 1.09] [-1.68, 0.67]

+ [2.5, -2.1]

+ [2.5, -2.1]

[4.37, -1.01]

[0.82, -1.43]

## Decoder

### Attention calculation in decoder (EOS)

EOS [2.70, -1.34] + positional encoding [0.0, 1.0] = [2.70, -0.34]

Q=0.4 0.4  
-0.3 0.1

K = 0.4 -0.7  
-0.4 0.3

V = -1.1 -0.7  
-0.4 1.3

QE = EOSXQ KE = EOSXK VE = EOSXV

QE = [1, 2, 1.0] KE = [1.2, -1.8] VE = [-2.8, -2.3]

QEXKE = -0.4 Softmax(-0.4) = 1.0

EOS[2.70, -0.34] + VE X 1.0 = [-0.1, -2.6].....(2)

$$PE(pos, 2i) = \sin \frac{pos}{10000^{\frac{2i}{d_{model}}}}$$

$$PE(pos, \frac{2i}{+1}) = \cos \frac{pos}{10000^{\frac{2i}{d_{model}}}}$$

## Decoder Encoder cross attention calculation (EOS, Lets, go)

$$Q = \begin{bmatrix} 1.5 & -0.3 \\ 0.3 & -1.0 \end{bmatrix} \quad K = \begin{bmatrix} -1.1 & 0.3 \\ -0.3 & -0.8 \end{bmatrix} \quad V = \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & -0.5 \end{bmatrix}$$

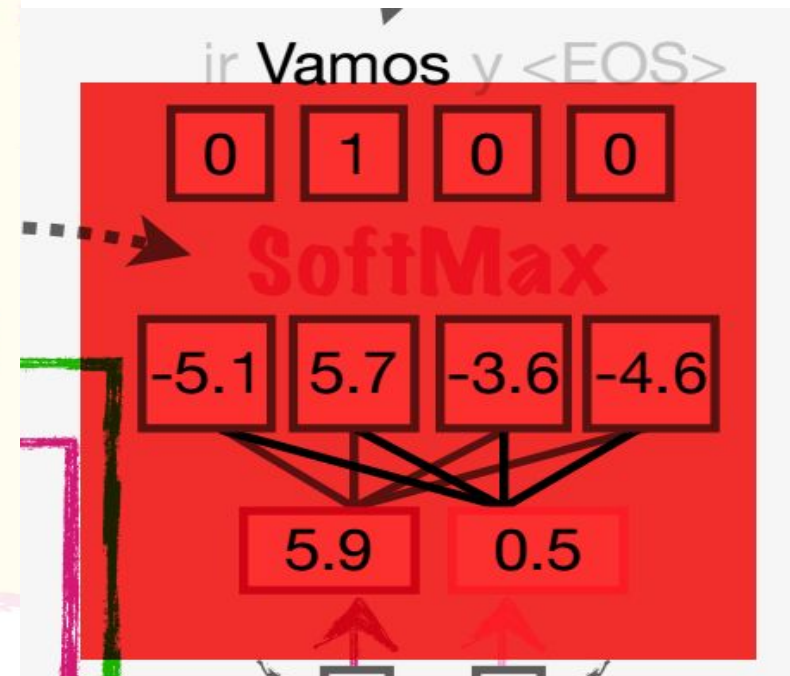
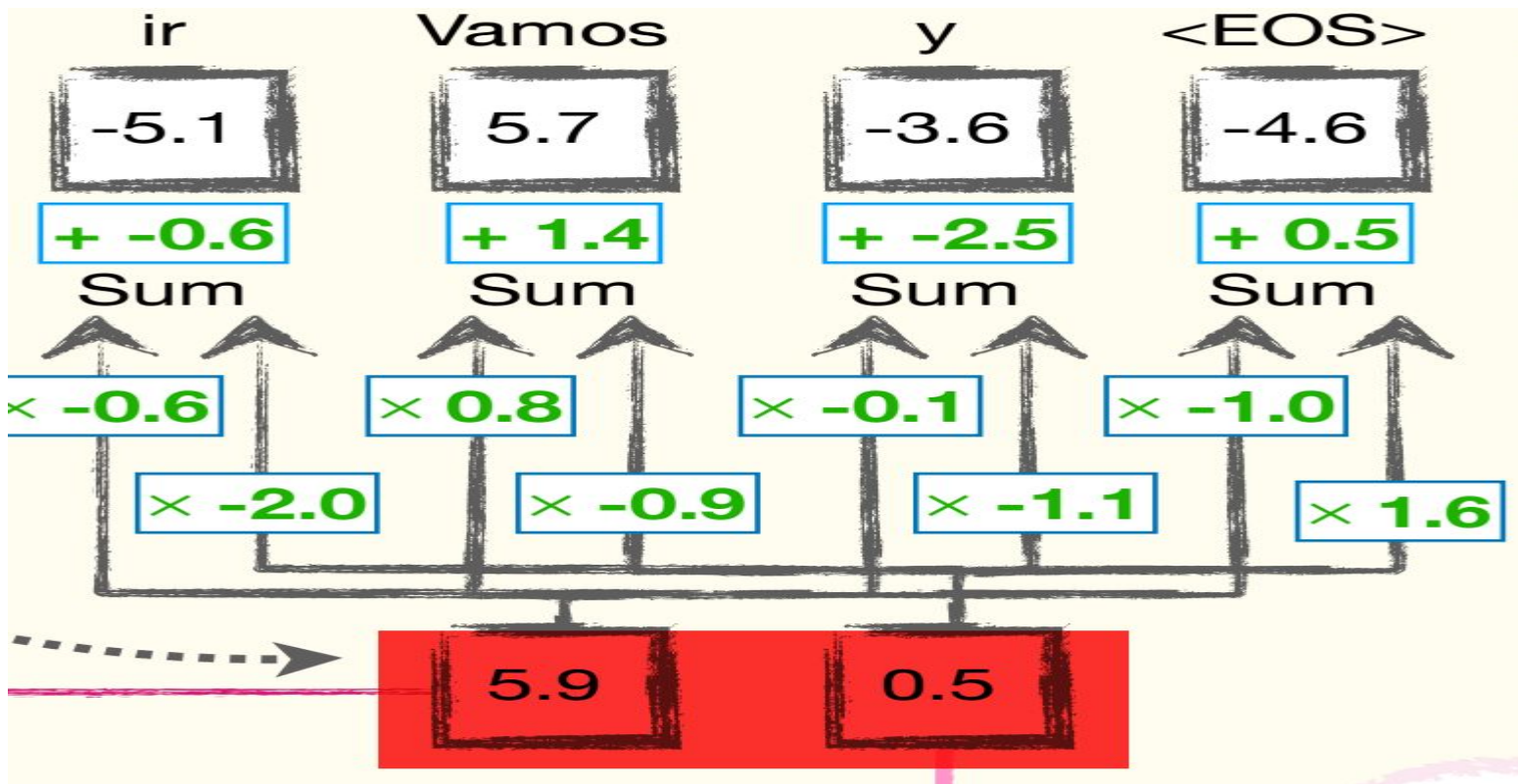
$$QE = \text{EOS} \times Q \quad KL = \text{Lets} \times K \quad KG = \text{go} \times K \\ VL = \text{Lets} \times V \quad VG = \text{go} \times V$$

$$QE = [-0.9, 2.6] \quad KL = [-4.5, 2.1] \quad VL = [6.0, 3.1] \\ KG = [-0.5, 1.4] \quad VG = [2.6, 1.2]$$

$$QExKL = 9.5 \quad QExKG = 4.1 \\ \text{Softmax}(9.5, 4.1) \\ 1.0, 0.0 \\ VL \times 1.0 + VG \times 0 \\ [6.0, 3.1]$$

$$\text{From (2) EOS } [-0.1, -2.6] + [6.0, 3.1] = [5.9, 0.5] \dots\dots\dots(3)$$

Generating first token in decoder after processing EOS



## Decoder

### Attention calculation in decoder (Vamos, EOS)

Vamos  $[-0.86, 2.37]$  + positional encoding  $[-0.9, 0.4] = [-0.86, 2.37]$

$Q = \begin{bmatrix} 0.4 & 0.4 \\ -0.3 & 0.1 \end{bmatrix}$ 
 $K = \begin{bmatrix} 0.4 & -0.7 \\ -0.4 & 0.3 \end{bmatrix}$ 
 $V = \begin{bmatrix} -1.1 & -0.7 \\ -0.4 & 1.3 \end{bmatrix}$

$KE = EOS \times K$      $VE = EOS \times V$      $QV = \text{Vamos} \times Q$      $VV = \text{vamos} \times V$   
 $KE = [1.2, -1.8]$      $VE = [-2.8, -2.3]$      $QV = [-1.3, -0.1]$      $VV = [0.0, 3.7]$

$QV \times KE = -1.1$      $QV \times KV = 1.4$   
 $\text{Softmax}(-1.1, -1.4) = 0.1, 0.9$   
 $\text{varmos}[-0.86, 2.37] + VE \times 0.1 + VV \times 0.9 = [-1.2, 5.5]$   
 .....(4)

### Decoder Encoder cross attention (Vamos, Lets, go)

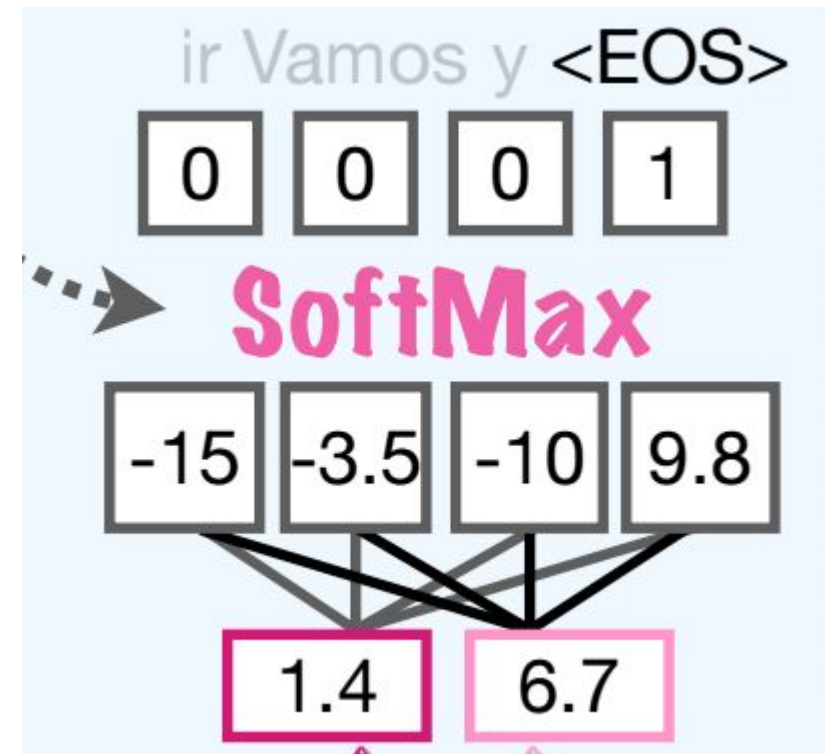
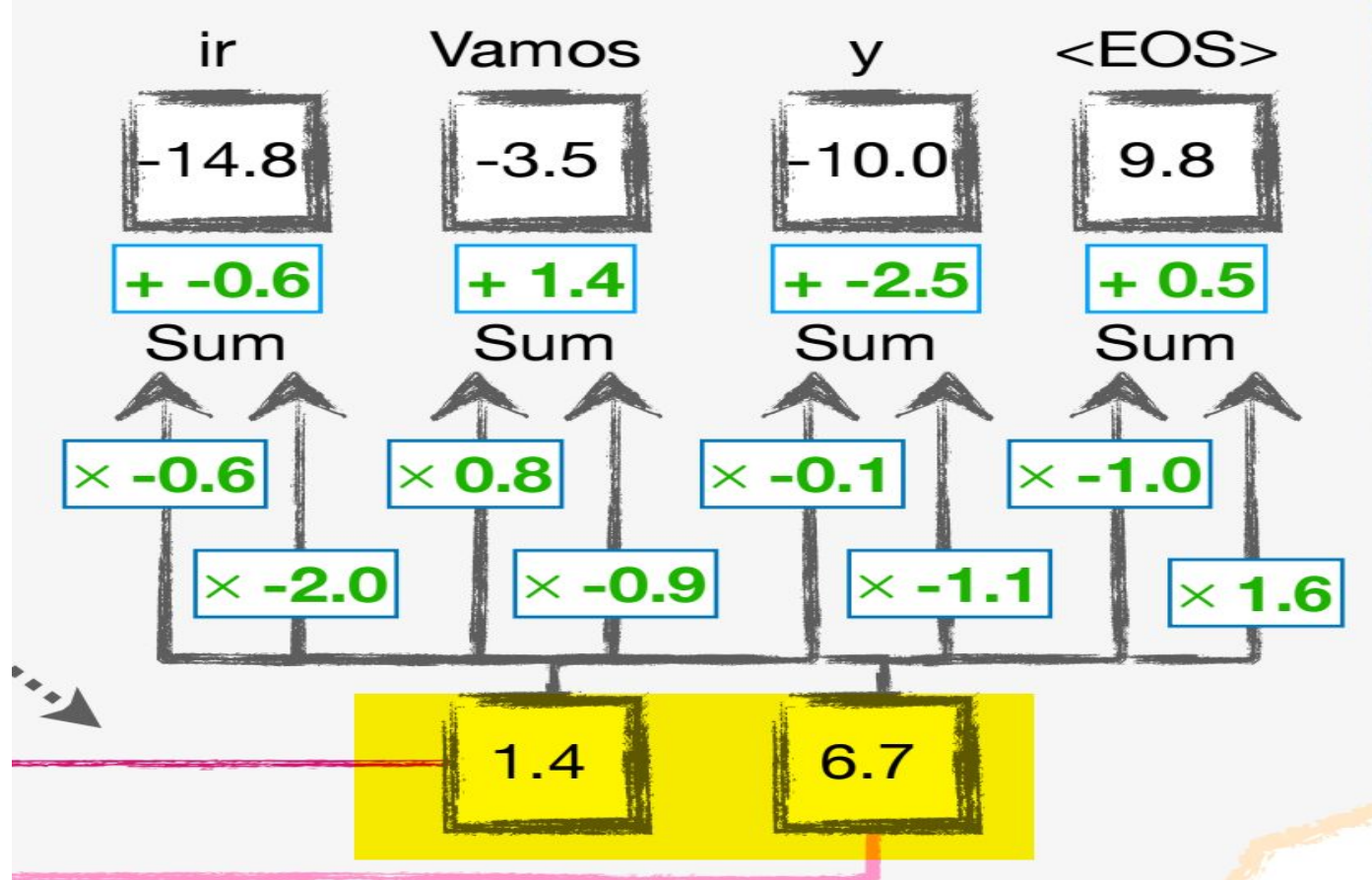
$Q = \begin{bmatrix} 1.5 & -0.3 \\ 0.3 & -1.0 \end{bmatrix}$ 
 $K = \begin{bmatrix} -1.1 & 0.3 \\ -0.3 & -0.8 \end{bmatrix}$ 
 $V = \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & -0.5 \end{bmatrix}$

$QV = \text{vamos} \times Q$      $KL = \text{Lets} \times K$      $KG = \text{go} \times K$   
 $VL = \text{Lets} \times V$      $VG = \text{go} \times V$

$QV = [-0.1, -5.1]$      $KL = [-4.5, 2.1]$      $VL = [6.0, 3.1]$   
 $KG = [-0.5, 1.4]$      $VG = [2.6, 1.2]$

$QV \times KL = -10.3$      $QV \times KG = -7.1$   
 $\text{Softmax}(-10.3, -7.1)$   
 $0.0, 1.0$   
 $VL \times 0.0 + VG \times 1.0$   
 $[2.6, 1.2]$

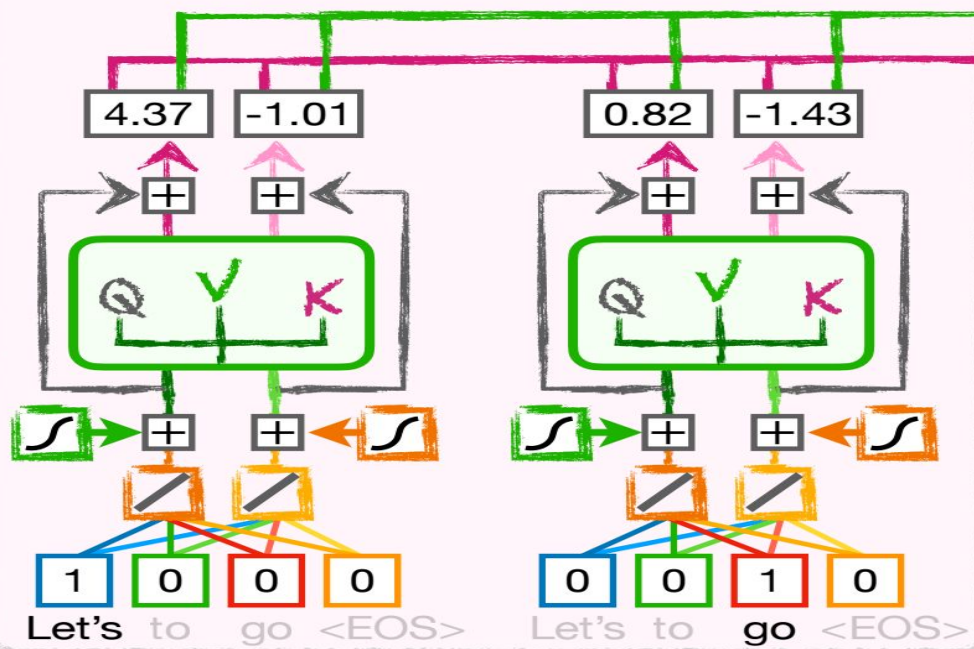
From (4) Vamos  $[-1.2, 5.5] + [2.6, 1.2] = [1.4, 6.7]$   
 .....(5)





and we're done decoding.

## Encoder



## Decoder

